14.8.1. GSM (Global System for Mobile Communication)

The Global System for Mobile Communications (GSM) is a second generation (2G) standard for mobile networks.



Figure 14.21 GSM Block Diagram

In the early 1980s, a group was formed by the European Telecommunications Standards Institute (ETSI) to develop a digital mobile communication system. Aptly named Groupe Speciale Mobile (GSM), its main task was to develop a single, consistent network for all of Europe and come up with a better and more efficient technical solution for wireless communication.



The GSM standard operates on three different carrier frequencies: the 900 MHz band, which was used by the original GSM system; the 1800 MHz band, which was added to support the swelling number of subscribers and the 1900 MHz frequency, which is used mainly in the U.S

CDMA (Code Division Multiple Access)

CDMA (Code-Division Multiple Access) refers to a protocol used in second-generation (2G) and third-generation (3G) wireless communications. Code Division Multiple Access (CDMA) is a sort of multiplexing that facilitates various signals to occupy a single transmission channel. It optimizes the use of available bandwidth. This technology is commonly used in ultra-high-frequency (UHF) cellular telephone systems, bands ranging between the 800-MHz and 1.9-GHz.

GPRS (General Packet Radio Service)

General Packet Radio Service (GPRS) is a packet-switching communications protocol for cellular networks. Mobile phones that have the GPRS technology can be used to receive data and information, such as web pages and email. GPRS is an "always on" technology which means that the mobile phone is always ready to receive data.